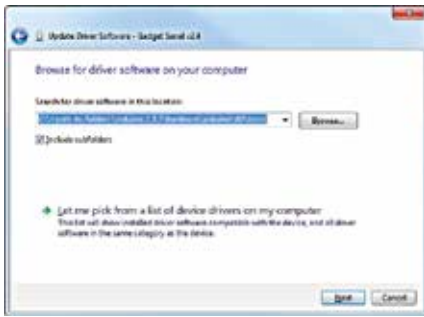


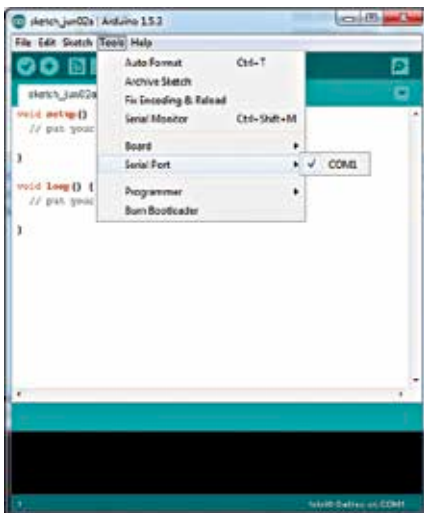


Path for Device Driver Installation

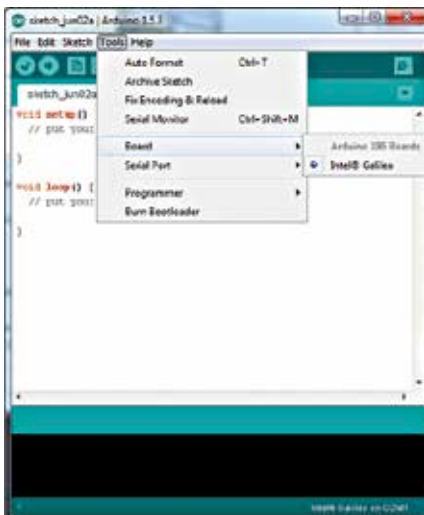
UPDATING FIRMWARE

It's better to update the Galileo board firmware after connecting it to your PC for the first time.

1. Power down the board and remove the SD card (if any). Again power up the board.
2. Open the Arduino IDE for Galileo.



Serial Port Selection in Arduino



Galileo Board Selection in Arduino



Galileo Firmware Upgrade (Serial Port Selection)

- a. Select the correct serial port (COM port) for your Galileo board under 'Tools' > 'Serial Port'. (COM1, in our case)
 - b. Select 'Galileo' board under 'Tools' > 'Board'
3. Check for confirmation at bottom-right corner.
 4. Navigate to 'Help' > 'Firmware Update'. Click 'Yes' to complete the updation process.
 5. Do not unplug the board or Disconnect the USB until the updating procedure is complete and a window saying 'Target Firmware Upgraded Successfully' pops up.

HARDWARE PROGRAMMING: THE SMALL DIFFERENCE

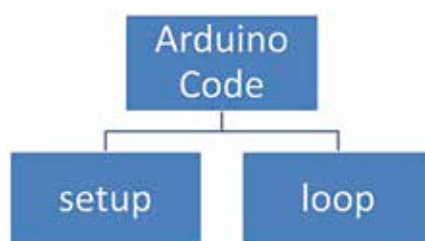
For this part, we assume you have slight knowledge of C/C++. E.g. Knowledge about functions.

PROCEDURE FOR WRITING CODE FOR HARDWARE DEVICE

Writing code for hardware devices is slightly different from writing software. It is similar to writing a driver for your hardware, where pin numbers need to be considered.

The biggest bug can show up due to a mistake when writing code for hardware devices, i.e. forgetting to run the code in a loop.

Arduino code consists of two parts (functions):



Division of Arduino code

PROGRAMMING THE GALILEO BOARD ARDUINO CODE

As in C/C++ anything between `/*` and `*/` and after `/**` is a comment.

Just as the entry point for a C program is the `main()` function, the entry point for Arduino code is the `setup()` function. Preprocessor directives can be used in Arduino code too.

Open the Arduino IDE for your platform and write the following program into the IDE.

Blink (Hello World!)

The program will blink an LED and is the electronics version of the 'Hello World!' program.

All Arduino compatible boards have an LED connected to their 13th pin that is used for preliminary testing purposes.

The code can be found under 'File' > 'Examples' > '01.Basics' > 'Blink'

Given below is the flowchart of the code:

DECIPHERING THE CODE ARDUINO + LINUX CODE

This example uses Linux system calls to create a Python script, which writes number 0-9 to a file called 'log.txt', one number per second. It then executes the Python script in the background, and reads the contents of the log file in the sketch while the Python script is updating it.

Linux commands can be used in Arduino using the `system()` function.

To run the 'ls' command that lists the files in directory, the Arduino code can be written as `system('ls')`.

Given below is the flowchart of the code that combines Linux Shell Programming:

DECIPHERING THE ARDUINO + LINUX CODE

More example codes and a Getting Started guide for Galileo board can be obtained at <http://dgit.in/inteldoc22192>.

LINUX ON GALILEO

The Galileo board has the advantage of being able to run Linux environment on board along with in the Arduino code, combining the best of both world.

The flash memory on board the Galileo is barely enough to store its Linux-based Yocto OS. The default Linux image has a few extra features added to it. The Linux OS can be loaded to a SD card, and Galileo can be set to boot from the Linux image. Using the Linux image, the following features become available: